EDS 223 Homework Assignment 3

Prioritizing potential aquaculture

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Read in data

```
library(sf)          # vector handling
library(terra)       # raster handling
library(tidyverse)
library(tmap)        # map making
library(kableExtra) # table formatting
```

```
sst2008 <- rast('data/average_annual_sst_2008.tif')
sst2009 <- rast('data/average_annual_sst_2009.tif')
sst2010 <- rast('data/average_annual_sst_2010.tif')
sst2011 <- rast('data/average_annual_sst_2011.tif')
sst2012 <- rast('data/average_annual_sst_2012.tif')
depth <- rast('data/depth.tif')
regions <- st_read('data/wc_regions_clean.shp')
```

```
# stack sst
sst <- c(sst2008, sst2009, sst2010, sst2011, sst2012)
```

Process data

Find the mean SST from 2008-2012 (e.g. create single raster of average SST)

```
mean_sst_k <- app(sst, fun = 'mean')
summary(mean_sst_k)
```

```
      mean
 Min.   :282.1
 1st Qu.:285.6
 Median :287.2
 Mean   :287.3
 3rd Qu.:289.2
 Max.   :301.2
 NAs    :42347
```

Convert average SST from Kelvin to Celsius (hint: subtract by 273.15)

```
mean_sst <- mean_sst_k - 273
summary(mean_sst)
```

```
      mean
 Min.   : 9.082
 1st Qu.:12.552
 Median :14.241
 Mean   :14.323
 3rd Qu.:16.214
 Max.   :28.200
 NAs    :42347
```

Crop depth raster to match the extent of the SST raster

- note: the resolutions of the SST and depth data do not match

```
depth_crop <- crop(depth, ext(mean_sst))
depth_crop <- project(depth_crop, mean_sst)
```

Resample the depth data to match the resolution of the SST data using the bilinear method approach

```
depth_crop_resamp <- resample(depth_crop, mean_sst, method = 'bilinear')
```

Check that the depth and SST match in resolution, extent, and coordinate reference system

- hint: can the rasters be stacked?

```
# confirm extent matches
cat("Extent matches:", ext(depth_crop_resamp) == ext(mean_sst))
```

```
Extent matches: TRUE
```

```
cat("Resolution matches:", res(depth_crop_resamp) == res(mean_sst))
```

```
Resolution matches: TRUE TRUE
```

```
cat("CRS matches:", same.crs(depth_crop_resamp, mean_sst))
```

```
CRS matches: TRUE
```

Find suitable locations

```
# stack mean_sst and depth
mean_sst_depth <- c(mean_sst, depth_crop_resamp)
```

```
map_habitat <- function(mintemp, maxtemp, mindepth, maxdepth,name){
  # define reclassify for sst
  rcl_sst <- matrix(c(
    -Inf, mintemp, 0,
    mintemp, maxtemp, 1,
    maxtemp, Inf, 0
  ), ncol = 3, byrow = TRUE)

  # define reclassify for oyster depth
  rcl_depth <- matrix(c(
    -Inf, mindepth, 0,
    mindepth, maxdepth, 1,
    maxdepth, Inf, 0
  ), ncol = 3, byrow = TRUE)
```

```
  sst_zones <- as.factor(classify(mean_sst_depth[[1]], rcl = rcl_sst))
  depth_zones <- as.factor(classify(mean_sst_depth[[2]], rcl = rcl_depth))

  # Find suitable habitats
  habitat_zone = as.factor(sst_zones * depth_zones)

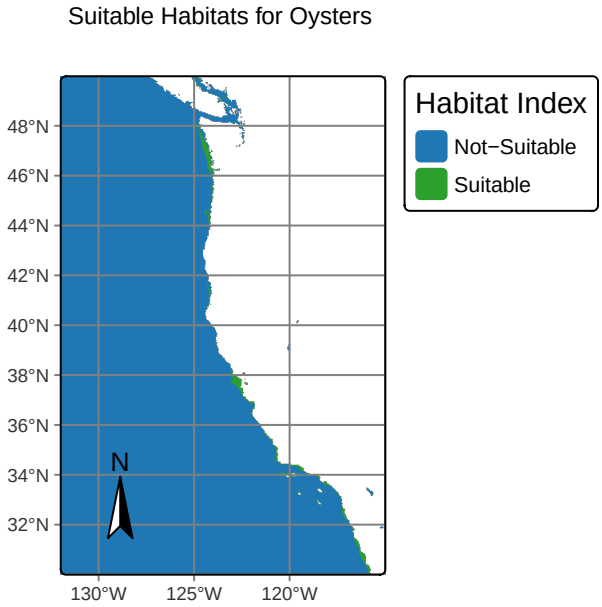
  levels(habitat_zone) <- data.frame(
    value = 0:1,
    label = c("Not-Suitable", "Suitable")
  )

  # Make map
  map <- tm_shape(habitat_zone) +
    tm_raster(col.legend = tm_legend("Habitat Index",
              position = tm_pos_out("right","center"))) +
    tm_title(paste("Suitable Habitats for",name)) +
    tm_graticules() +
    tm_compass(position = c('bottom','left'))

  # Determine most suitable EEZ
  hab_area <- cellSize(habitat_zone, unit = 'km') |>
    mask(habitat_zone, maskvalues = 0)
  habitable_area <- zonal(hab_area, vect(regions), fun = 'sum', na.rm = TRUE)
  count_by_zone <- zonal(habitat_zone, vect(regions), fun = "sum", na.rm = TRUE)
  count_by_region <- cbind(region = regions$rgn,
                           count = count_by_zone,
                           area_km2 = habitable_area)
  colnames(count_by_region) <- c('Region', 'Count','Area(km^2)')

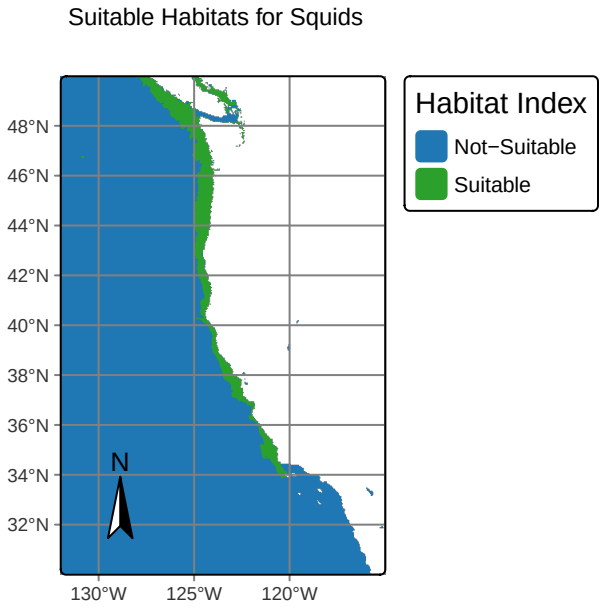
  # Display the outputs
  print(map)
  print(count_by_region)
}
```

```
# Make map of oysters
map_habitat(11,30,-70,0,"Oysters")
```



	Region	Count	Area(km^2)
1	Oregon	89	2547.540
2	Northern California	28	1140.895
3	Central California	218	4988.420
4	Southern California	172	3775.949
5	Washington	219	6226.850

```
# Make map of the common squid
map_habitat(10,14,-700,0,'Squids')
```



	Region	Count	Area(km^2)
1	Oregon	1650	26540.119
2	Northern California	746	12922.949
3	Central California	921	17123.111
4	Southern California	249	5119.809
5	Washington	1124	19358.250

Reflection

Looking at the maps, oysters have a small sliver of habitable areas. Oysters have a predominantly suitable habitat in Washington, then Central California, with Northern California having the least area for suitable habitats.

For squids, the area for suitable habitats are larger, with Oregon and Washington having the top two largest area. Southern California has the smallest area for suitable habitats for squids.